INDIAN INSTITUTE OF TECHNOLOGY HYDERABAD

DEPARTMENT OF MATHEMATICS

End Semester Examination — Time : 120 minutes

Date : 20.03.2026 — MA 2130 : Complex Variables — Max Marks: 40

Reg. Number & Name: EE24BTECH11203 ANOOP K

Answer all parts of the same question at one place. All solutions should have a valid mathematical justification

Question:	1	2	3	4	5	6	7	8	9	10	Total
Points:	4	4	4	4	4	4	4	4	4	4	40
Score:											

Justify your solutions:

1. (a) Find all the values of $z \in \mathbb{C}$ such that $z^5 = -32$. [2]

   (b) What is the value of the integral $\int_0^{2\pi} \sin^2(\frac{\pi}{6} + 2e^{i\theta})d\theta$? [2]

2. If $a > e$, then prove that the equation $az^n = e^z$ has n roots inside the contour $|z| = 1$. [4]

3. Prove that the function $u(x,y) = 3x^2y + 2x^2 - y^3 - 2y^2$ is harmonic. Find a function $v(x,y)$ such that $f(z) = u(x,y) + iv(x,y)$, where $z = x + iy$, is analytic. [4]

4. (a) Write down all the singularities (with multiplicity) of the function [2]

$$f(z) = \frac{(z+3i)^2}{(z^2-2z+5)^2}.$$

   (b) Determine all the analytic functions $f : \mathbb{C} \to \mathbb{C}$ satisfying $f(z) = f(z+1) = f(z+i)$ for all $z \in \mathbb{C}$. [2]

5. Using methods from complex variable theory, evaluate the integral $\int_0^{2\pi} \frac{\cos 3\theta}{5 - 4\cos\theta} d\theta$. [4]

6. (a) Let $z \in \mathbb{C} \setminus \{0\}$ and $c \in \mathbb{C}$. Define $z^c := e^{c \log(z)}$ and the principal value of $z^c := e^{c \operatorname{Log}(z)}$. Determine all values of i^{-2i} and the principal value of i^{-2i}. [2]

   (b) Write down the Laurent series expansion of the function $f(z) = \frac{1}{(z-1)(z+1)}$ in the annulus $1 < |z| < 2$. [2]

7. Define an isolated singularity of a function $f(z)$. Calculate the residue of $\frac{\operatorname{Log}(z)}{(z^2+1)^2}$ at all isolated singular points. [4]

8. Let $f(z)$ be an analytic function in the region $|z| \leq R$. Suppose that $|f(z)| \leq M$ in the region $|z| \leq R$. If $f(0) = 0$, then show that [4]

$$|f(z)| \leq \frac{M|z|}{R}$$

for all $z \in \mathbb{C}$ with $|z| \leq R$.

9. Using methods from complex variable theory, evaluate the integral [4]

$$\int_0^\infty \frac{x^2}{(x^2+1)(x^2+4)} dx.$$

10. Define an anti-derivative of a function $f(z)$ on a domain D. Does the function [4]

$$f(z) = \frac{\overline{z}}{(1+|z|^2)^2}$$

admit an anti-derivative on $\mathbb{C}$?